Manual FAS Calibration Procedure

- Script to Generate Guide
- Meaurement Guide
- Warning: SDBB
- General Guide
  - 1. Measurement Data & Conversion
    - Quick Conversion Reference
    - Command Data Structure
  - 2. FLCW FAS Calibration (Front Camera)
    - Step 1: Security Access (HPA)
    - Step 2: Write Alignment Data
    - Step 3: Verify & Calibrate
    - Step 4: Post-Calibration Checks
  - 3. SRCx FAS Calibration (Surround View)
    - Step 1: Security Access (HPA)
    - Step 2: Write Alignment Data
    - Step 3: Verify & Calibrate
    - Step 4: Post-Calibration Checks
  - 4. LRBx FAS Calibration (Rear Side Cameras)
    - Step 1: Security Access (HPB)
    - Step 2: Write Alignment Data
    - Step 3: Verify & Calibrate
    - Step 4: Post-Calibration Checks
  - 5. LRIx FAS Calibration
  - Quick Reference
    - Security Access Summary
    - Command Summary Table
    - Alignment Data Command Structure
    - DTC Status Reference
    - Common NRC Responses
  - Notes

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Script to Generate Guide

Run the script, it will ask u to input the measurement data, after that it will generate a markdown for u, then u only need to follow the generated markdown file, copy, paste, then done :))

If u don't know/don't want to use the py, below is the detailed guide, modified based on the script generated file

Meaurement Guide

4. Wheel Arch Height Measurement

Warning: SDBB

Plz make sure you have the matching SDBB file for the calibration. For REL12, use 726A with HPA/HPB added.

General Guide

1. Measurement Data & Conversion

Before sending commands, convert your wheel arch height measurements (mm) to 4-digit Hex.

Quick Conversion Reference

Measurement (mm)	Hex Value	4-Digit Hex
828	33C	`033C`
830	33E	`033E`
850	352	`0352`

Command Data Structure

The alignment data is always appended in this order: `[FL] [FR] [RL] [RR]`.

2. FLCW FAS Calibration (Front Camera)

DID: `4603`

Step 1: Security Access (HPA)

1. **Enter Extended Session:** `10 03`
2. **Unlock Security Level 17:** `27 17` → (Key calc) → `27 18 ...`
3. **Disable Firewall:** `2E D0 74 00` *Note: If the car has no firewall, this may return* `NRC 31` *(RequestOutOfRange). This is acceptable, proceed.*
4. **Unlock Security Level 05:** `27 05` → (Key calc) → `27 06 ...`

Step 2: Write Alignment Data

Structure: `2E 46 03 0B DA 05 78 [FL] [FR] [RL] [RR]`

Example (FL:830, FR:828, RL:830, RR:830): `2E 46 03 0B DA 05 78 03 3E 03 3C 03 3E 03 3E (Put a 0 in front of every hex values)`

Step 3: Verify & Calibrate

1. **Confirm Data Written:** `22 46 03`
2. **Start Calibration Routine:** `31 01 21 21`
3. **Check Routine Results:** `31 03 21 21` *(Look for* `00` *or success status)*

Step 4: Post-Calibration Checks

1. **Read Calibration Result DID:** `22 46 66`
2. **Read Intrinsic Parameters:** `22 46 7D`
3. **Verify Files (WinSCP):**
   - `/var/zenseact`
   - `/var/fs/zen_qm_vision`

3. SRCx FAS Calibration (Surround View)

DID: `4604`

Step 1: Security Access (HPA)

1. **Enter Extended Session:** `10 03`
2. **Unlock Security Level 17:** `27 17` ...
3. **Disable Firewall:** `2E D0 74 00`

4. **Unlock Security Level 05:** `27 05` ...

Step 2: Write Alignment Data

Structure: `2E 46 04 [FL] [FR] [RL] [RR]`

Example (FL:830, FR:828, RL:830, RR:830): `2E 46 04 03 3E 03 3C 03 3E 03 3E`

Step 3: Verify & Calibrate

1. **Confirm Data Written:** `22 46 04`
2. **Start Calibration Routine:** `31 01 21 22`
3. **Check Routine Results:** `31 03 21 22`

Step 4: Post-Calibration Checks

1. **Read Calibration Result DID:** `22 46 7E`
2. **Read Intrinsic Parameters:** `22 46 7F`
3. **Verify Files (WinSCP):**
   - `/var/fs/zen_sense_a`

4. LRBx FAS Calibration (Rear Side Cameras)

DID: `4734` (Note: This is usually on **HPB**, requiring separate connection/security).

Step 1: Security Access (HPB)

1. **Enter Extended Session:** `10 03`
2. **Unlock Security Level 17:** `27 17` ...
3. **Disable Firewall:** `2E D0 74 00`
4. **Unlock Security Level 05:** `27 05` ...

Step 2: Write Alignment Data

Structure: `2E 47 34 13 39 09 59 04 4C 41 9C 00 00 [FL] [FR] [RL] [RR]`

Example (FL:830, FR:828, RL:830, RR:830): `2E 47 34 13 39 09 59 04 4C 41 9C 00 00 03 3E 03 3C 03 3E 03 3E`

Step 3: Verify & Calibrate

1. **Confirm Data Written:** `22 47 34`
2. **Start Calibration Routine:** `31 01 21 23`
3. **Check Routine Results:** `31 03 21 23`

Step 4: Post-Calibration Checks

1. **Read Calibration Result DID:** `22 47 35`
2. **Read Intrinsic Parameters:** `22 47 36`
3. **Verify Files (WinSCP):**
   - `/var/fs/zen_sense_b`

5. LRIx FAS Calibration

Step 1: Security Access

Enter Extended Session: 10 03

Unlock Security Level 17: 27 17 ...

Disable Firewall: 2E D0 74 00

Unlock Security Level 05: 27 05 ...

Step 2: Write Alignment Data

Structure: 2E 4B 8E 03 83 09 EB 03 52 00 00 00 00 [FL] [FR] [RL] [RR]

Step 3: Verify & Calibrate

Confirm Data Written: 22 4B 8E

Start Calibration Routine: 31 01 02 6E

Check Routine Results: 31 03 02 6E

Step 4: Post-Calibration Checks

Read Calibration Result DID: 22 4A 92

Read Intrinsic Parameters: 22 4B 8F

Note:

Save all the verified (WinSCP) files

Quick Reference

Security Access Summary

ECU	Security Level	Method
HPA	17 & 05	17 Automatic via DSA file transfer; 95 need to manually unlock
HPB	05	Manual unlock required

Command Summary Table

Camera	Alignment Data DID	Start Routine	Result Routine	Result DID	Intrinsic DID
FLCW	4603	`31 01 2121`	`31 03 2121`	`22 4666`	`22 467D`
SRCx	4604	`31 01 2122`	`31 03 2122`	`22 467E`	`22 467F`
LRBx	4734	`31 01 2123`	`31 03 2123`	`22 4735`	`22 4736`

Alignment Data Command Structure

Camera	Command Structure
FLCW	`2E 4603 0B DA 05 78[FL] [FR] [RL] [RR]`
SRCx	`2E 4604 [FL] [FR] [RL] [RR]`
LRBx	2E 4734 13 39 09 59 04 4C 41 9C 00 00 xFLx xFRx xRLx xRRx

DTC Status Reference

Status	Description
2F	Test Failed (Active)
50	Test Passed (Inactive)
00	Test Not Completed
2E	Test Completed - Passed This Cycle

Common NRC Responses

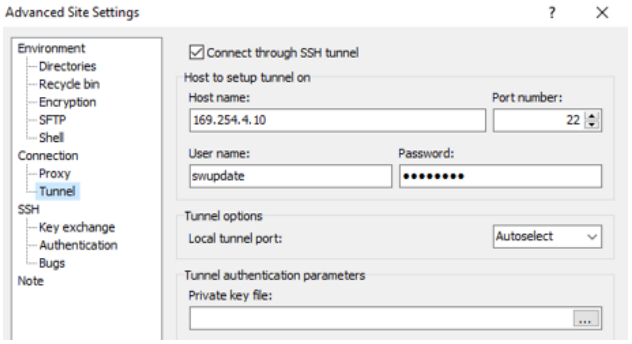
NRC	Meaning	Possible Cause
0x10	General Reject	System not ready
0x22	Conditions Not Correct	Camera info not received or can't acquire image
0x31	Request Out of Range	Invalid DID/RID or not registered

Notes

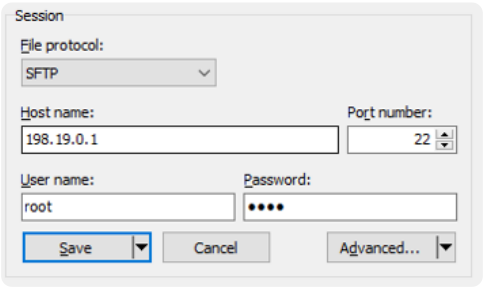
- **HPA Security:** Automatically handled by DSA file transfer (no manual unlock needed)
- **HPB Security:** Manual unlock required before calibration
- **Measurement Range:** Typical values are 500-1200mm
- **Hex Conversion:** Use Windows Calculator (Programmer mode) or online hex converter
- **Command Order:** Alignment data is always in order: Front Left, Front Right, Rear Left, Rear Right
- **File Verification:** Always verify calibration files are created before proceeding
- **DTC Verification:** All DTCs should show "Passed" status (50/00/2E) after successful calibration

WINSCP

- Configure SSH Tunnel (Password is "swupdate")
- :

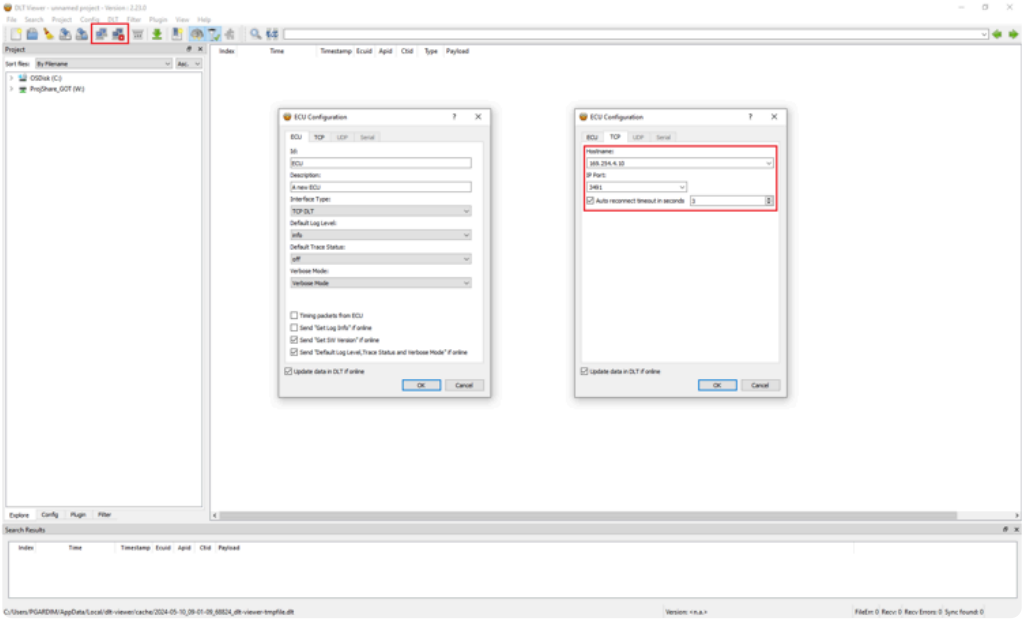


- For HPA (Password is "root"):
-



- Post Calibration

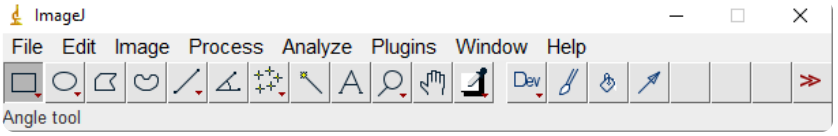
DLT Logging

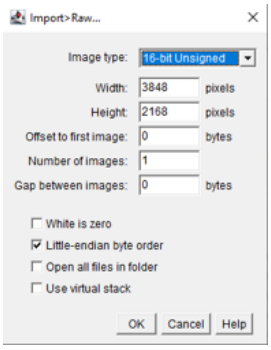


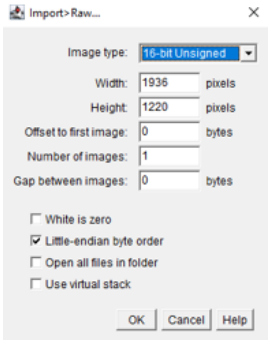
  - Configure DLT logging according to the "ECU" and "TCP" tabs.
  - Start DLT by pressing the first button marked on the top.
  - Stop DLT by pressing the second button marked on the top.

Drag the .raw files from /var/log/csp:

- zen_qm_vision_factory_calibration_camera_flcw.dump
  zen_qm_vision_factory_calibration_camera_flcw.raw
  zen_qm_vision_factory_calibration_camera_srcb.dump
  zen_qm_vision_factory_calibration_camera_srcb.raw
  zen_qm_vision_factory_calibration_camera_srcf.dump
  zen_qm_vision_factory_calibration_camera_srcf.raw
  zen_qm_vision_factory_calibration_camera_srcl.dump
  zen_qm_vision_factory_calibration_camera_srcl.raw
  zen_qm_vision_factory_calibration_camera_srcr.dump
  zen_qm_vision_factory_calibration_camera_srcr.raw

- Use ImageJ tool
  - 

- Drag and paste the .raw files to any place in the SW
  - In the pop-up that will show up, configure it as follow:
    - 

FLCW / LRBx / FLCL: "16-bit Unsigned", 3848, 2168, Little-endian byte order.
    - 

SRCx: "16-bit Unsigned", 1936, 1220, Little-endian byte order.
  - Press OK, and the image will show up.

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